



National Textile University
Department of Computer Science

Subject:
Operating System

Submitted to:
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Submitted by:
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Reg number:
23-NTU-CS-1126

Lab no:
3 - hometask

Semester:
5th

PART 1

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ mkdir Lab_3/docs/drafts  
afiamaham@DESKTOP-190QPQH:~$ ls docs  
ls: cannot access 'docs': No such file or directory  
afiamaham@DESKTOP-190QPQH:~$ ls Lab_3/docs  
drafts  
afiamaham@DESKTOP-190QPQH:~$ mkdir Lab_3/data/raw  
afiamaham@DESKTOP-190QPQH:~$ mkdir Lab_3/data/processed  
afiamaham@DESKTOP-190QPQH:~$ ls Lab_3/data  
processed raw  
afiamaham@DESKTOP-190QPQH:~$
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ touch Lab_3/docs/intro.txt  
afiamaham@DESKTOP-190QPQH:~$ touch Lab_3/docs/notes.txt  
afiamaham@DESKTOP-190QPQH:~$ touch Lab_3/docs/summary.txt  
afiamaham@DESKTOP-190QPQH:~$ ls Lab_3/docs  
drafts intro.txt notes.txt summary.txt  
afiamaham@DESKTOP-190QPQH:~$
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ echo "Inside intro.txt. It is a sample text.">Lab_  
afiamaham@DESKTOP-190QPQH:~$ cat Lab_3/docs/intro.txt  
Inside intro.txt. It is a sample text.  
afiamaham@DESKTOP-190QPQH:~$ echo "Inside notes.txt. It is a sample text of Lab  
afiamaham@DESKTOP-190QPQH:~$ cat Lab_3/docs/notes.txt  
Inside notes.txt. It is a sample text of Lab 3.  
afiamaham@DESKTOP-190QPQH:~$ echo "Random text in OS.">Lab_3/docs/summary.txt  
afiamaham@DESKTOP-190QPQH:~$ cat Lab_3/docs/summary.txt  
Random text in OS.  
afiamaham@DESKTOP-190QPQH:~$
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ cp Lab_3/docs/summary.txt Lab_3/docs/drafts  
afiamaham@DESKTOP-190QPQH:~$ ls Lab_3/docs/drafts  
summary.txt  
afiamaham@DESKTOP-190QPQH:~$
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ touch Lab_3/data/raw/raw1.txt  
afiamaham@DESKTOP-190QPQH:~$ touch Lab_3/data/raw/raw2.txt  
afiamaham@DESKTOP-190QPQH:~$ ls Lab_3/data/raw  
raw1.txt  raw2.txt  
afiamaham@DESKTOP-190QPQH:~$ _
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ date>> Lab_3/data/raw/raw1.txt  
afiamaham@DESKTOP-190QPQH:~$ cat Lab_3/data/raw/raw1.txt  
Tue Oct 7 15:20:34 PKT 2025  
afiamaham@DESKTOP-190QPQH:~$
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ mv Lab_3/data/raw/raw2.txt Lab_3/data/pro  
afiamaham@DESKTOP-190QPQH:~$ ls Lab_3/data/raw  
raw1.txt  
afiamaham@DESKTOP-190QPQH:~$ ls Lab_3/data/processed  
raw2.txt  
afiamaham@DESKTOP-190QPQH:~$
```

```
GNU nano 7.2 Lab_3/scripts/hello.sh  
echo "Hello world"  
pwd  
ls -lh_  
[ Wrote 3 lines ]  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute  
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

```
afiamaham@DESKTOP-190QPQH: ~$ ls -R
.:
Lab_3  'Operating System'  lab01  test

./Lab_3:
data  do  docs  scripts

./Lab_3/data:
processed  raw

./Lab_3/data/processed:
raw2.txt

./Lab_3/data/raw:
raw1.txt

./Lab_3/docs:
```

```
./Lab_3/data/raw:
raw1.txt

./Lab_3/docs:
drafts  intro.txt  notes.txt  summary.txt

./Lab_3/docs/drafts:
summary.txt

./Lab_3/scripts:
hello.sh

'./Operating System':
lab03-task01.c  lab03-task03.c  task01-output  task03-output
lab03-task02.c  lab03-task04.c  task02-output  task04-output
```

PART 2:

```
afiamaham@DESKTOP-190QPQH: ~/Lab_3
afiamaham@DESKTOP-190QPQH:~/Lab_3$ pwd
/home/afiamaham/Lab_3
afiamaham@DESKTOP-190QPQH:~/Lab_3$ whoami
afiamaham
afiamaham@DESKTOP-190QPQH:~/Lab_3$ touch extra.txt
afiamaham@DESKTOP-190QPQH:~/Lab_3$
```

```
afiamaham@DESKTOP-190QPQH: ~/Lab_3
afiamaham@DESKTOP-190QPQH:~/Lab_3$ pwd
/home/afiamaham/Lab_3
afiamaham@DESKTOP-190QPQH:~/Lab_3$ whoami
afiamaham
afiamaham@DESKTOP-190QPQH:~/Lab_3$ touch extra.txt
afiamaham@DESKTOP-190QPQH:~/Lab_3$ cat docs/intro.txt
Inside intro.txt. It is a sample text.
afiamaham@DESKTOP-190QPQH:~/Lab_3$ rm extra.txt
afiamaham@DESKTOP-190QPQH:~/Lab_3$ history | tail -n 5
264  whoami
265  touch extra.txt
266  cat docs/intro.txt
267  rm extra.txt
268  history | tail -n 5
afiamaham@DESKTOP-190QPQH:~/Lab_3$
```

PART 3:

```
afiamaham@DESKTOP-190QPQH: ~/Lab_3
afiamaham@DESKTOP-190QPQH:~/Lab_3$ chmod u=rwx,g=rwx,o=--- scripts/hello.sh
afiamaham@DESKTOP-190QPQH:~/Lab_3$ ls -l scripts/hello.sh
-rwxrwx--- 1 afiamaham afiamaham 30 Oct  7 15:27 scripts/hello.sh
afiamaham@DESKTOP-190QPQH:~/Lab_3$ ./scripts/hello.sh
Hello world
/home/afiamaham/Lab_3
total 16K
drwxr-xr-x 4 afiamaham afiamaham 4.0K Oct  7 15:00 data
-rw-r--r-- 1 afiamaham afiamaham  39 Oct  7 15:09 do
drwxr-xr-x 3 afiamaham afiamaham 4.0K Oct  7 15:02 docs
drwxr-xr-x 2 afiamaham afiamaham 4.0K Oct  7 15:27 scripts
afiamaham@DESKTOP-190QPQH:~/Lab_3$
```

```
afiamaham@DESKTOP-190QPQH: ~/Lab_3
afiamaham@DESKTOP-190QPQH:~/Lab_3$ chmod 664 docs/intro.txt
afiamaham@DESKTOP-190QPQH:~/Lab_3$ ls -l docs/intro.txt
-rw-rw-r-- 1 afiamaham afiamaham 39 Oct  7 15:10 docs/intro.txt
afiamaham@DESKTOP-190QPQH:~/Lab_3$
```

```
afiamaham@DESKTOP-19OQPQH: ~/Lab_3
afiamaham@DESKTOP-19OQPQH:~/Lab_3$ chmod 770 docs/notes.txt
afiamaham@DESKTOP-19OQPQH:~/Lab_3$ ls -l docs/notes.txt
-rwxrwx--- 1 afiamaham afiamaham 48 Oct  7 15:12 docs/notes.txt
afiamaham@DESKTOP-19OQPQH:~/Lab_3$
```

PART 4:

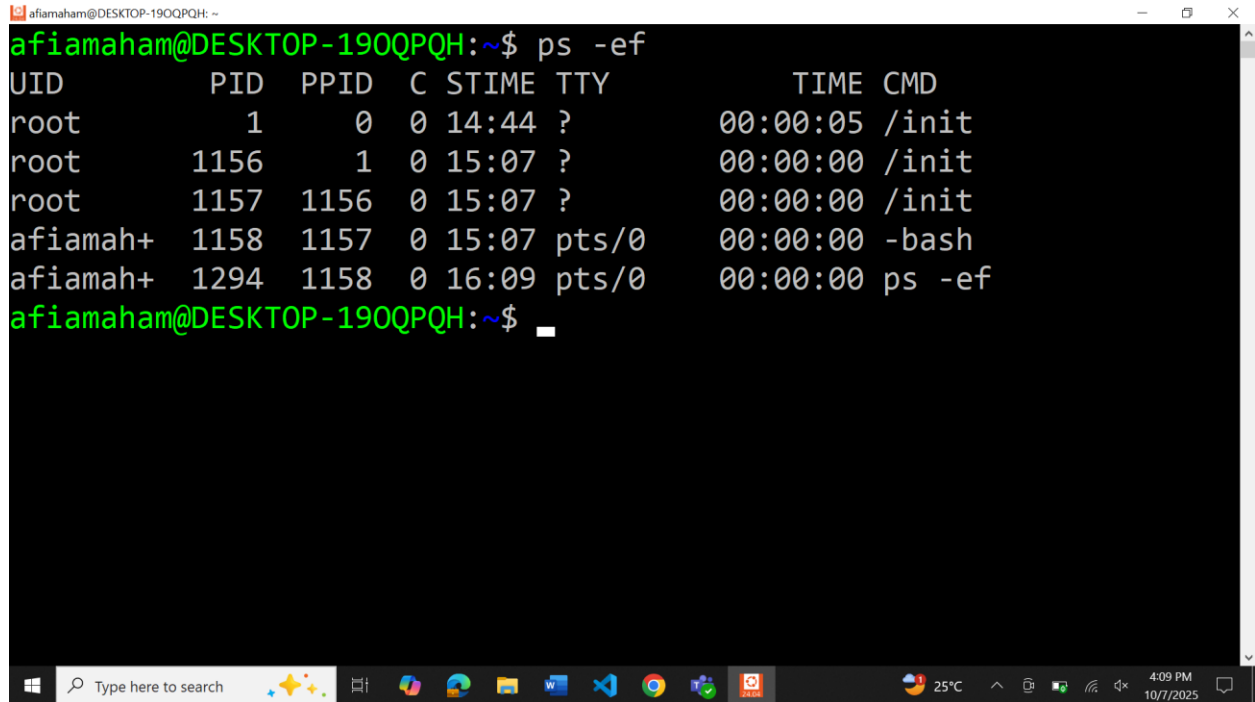
```
afiamaham@DESKTOP-19OQPQH: ~/Lab_3
afiamaham@DESKTOP-19OQPQH:~/Lab_3$ wc docs/notes.txt
 1 10 48 docs/notes.txt
afiamaham@DESKTOP-19OQPQH:~/Lab_3$ head -n 2 docs/summary.txt
Random text in OS.
It is a summary.txt file.
afiamaham@DESKTOP-19OQPQH:~/Lab_3$ tail -n 1 docs/summary.txt
We are writing in a file.
afiamaham@DESKTOP-19OQPQH:~/Lab_3$ grep "intro" docs/intro.txt
Inside intro.txt. It is a sample text.
afiamaham@DESKTOP-19OQPQH:~/Lab_3$
```

PART 5:

```
afiamaham@DESKTOP-190QPQH: ~$ ps -ef
```

UID	PID	PPID	C	STIME	TTY	TIME	CMD
root	1	0	0	14:44	?	00:00:05	/init
root	1156	1	0	15:07	?	00:00:00	/init
root	1157	1156	0	15:07	?	00:00:00	/init
afiamah+	1158	1157	0	15:07	pts/0	00:00:00	-bash
afiamah+	1294	1158	0	16:09	pts/0	00:00:00	ps -ef

```
afiamaham@DESKTOP-190QPQH: ~$
```



Process taking highest CPU and Memory

CPU: root 3.0

EMMI: afiamaham 0.1

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ yes > /dev/null &  
[1] 1298  
afiamaham@DESKTOP-190QPQH:~$ ps -ef | grep yes  
afiamah+ 1298 1158 99 16:12 pts/0 00:00:28 yes  
afiamah+ 1300 1158 0 16:13 pts/0 00:00:00 grep --color=auto y  
es  
afiamaham@DESKTOP-190QPQH:~$ kill 1298  
afiamaham@DESKTOP-190QPQH:~$ ps -ef | grep yes  
afiamah+ 1302 1158 0 16:13 pts/0 00:00:00 grep --color=auto y  
es  
[1]+ Terminated yes > /dev/null  
afiamaham@DESKTOP-190QPQH:~$
```

```
afiamaham@DESKTOP-190QPQH: ~  
afiamaham@DESKTOP-190QPQH:~$ sleep 60  
^C  
afiamaham@DESKTOP-190QPQH:~$ sleep 60 &  
[1] 1305  
afiamaham@DESKTOP-190QPQH:~$ fg  
sleep 60  
^Z  
[1]+ Stopped sleep 60  
afiamaham@DESKTOP-190QPQH:~$ bg  
[1]+ sleep 60 &  
afiamaham@DESKTOP-190QPQH:~$ jobs  
[1]+ Running sleep 60 &  
afiamaham@DESKTOP-190QPQH:~$
```

PART 6:

Program 1:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main() {
    pid_t pid;

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        exit(1);
    }
    else if (pid == 0) {
        printf("Child process (PID: %d) running top...\n", getpid());
        execlp("top", "top", NULL);
        perror("execlp failed");
    }
    else {
        printf("Parent process (PID: %d) created child with PID: %d\n", getpid(), pid);
        sleep(60);
        printf("Parent process finished.\n");
    }

    return 0;
}
```

```
Lab03-hometask > C task01.c
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4
5 int main() {
6     pid_t pid;
7
8     pid = fork();
9
10    if (pid < 0) {
11        perror("fork failed");
```

afiamaham@DESKTOP-190QPQH:~/Operating System\$ gcc Lab03-hometask/task01.c -o task01-output

afiamaham@DESKTOP-190QPQH:~/Operating System\$./task01-output

Parent process (PID: 2106) created child with PID: 2107

Child process (PID: 2107) running top...

top - 16:24:24 up 1:39, 0 user, load average: 0.01, 0.06, 0.04

Tasks: 19 total, 1 running, 18 sleeping, 0 stopped, 0 zombie

%Cpu(s): 0.9 us, 0.8 sy, 0.0 ni, 98.2 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st

Mem: 6247.3 total, 5750.9 free, 518.5 used, 193.7 buff/cache

Mem Swap: 2048.0 total, 2048.0 free, 0.0 used, 5728.9 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1496	afiamah+	20	0	31.9g	147588	53936	S	4.0	2.3	0:19.62	node
1434	afiamah+	20	0	11.3g	110964	51348	S	1.7	1.7	0:08.63	node
1447	afiamah+	20	0	1015208	56628	42448	S	0.7	0.9	0:01.13	node
1605	afiamah+	20	0	1064704	76888	47884	S	0.7	1.2	0:03.04	node
1446	root	20	0	1200	124	20	S	0.3	0.0	0:00.48	init
1	root	20	0	1200	476	476	S	0.0	0.0	0:05.74	init
1422	root	20	0	1200	124	20	S	0.0	0.0	0:00.00	init

```
afiamaham@DESKTOP-190QPQH:~/Operating System$ ps -ef | grep top
afiamah+ 2107 1423 0 16:24 pts/3 00:00:00 top
afiamah+ 3077 3030 0 16:27 pts/4 00:00:00 grep --color=auto top

afiamaham@DESKTOP-190QPQH:~/Operating System$ kill 2107

afiamaham@DESKTOP-190QPQH:~/Operating System$ ps -ef | grep top
afiamah+ 3197 3030 0 16:27 pts/4 00:00:00 grep --color=auto top

afiamaham@DESKTOP-190QPQH:~/Operating System$
```

Program 2:

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    pid_t pid = fork(); // create child process

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }
    else if (pid == 0) {
        // Child process
        printf("Child (PID: %d) running date command...\n", getpid());
        execlp("date", "date", NULL); // replace process with date command
        perror("execlp failed"); // runs only if exec fails
    }
    else {
        // Parent process
        printf("Parent (PID: %d) waiting for child (PID: %d)...\n", getpid(), pid);
        wait(NULL); // wait for child to finish
        printf("Child finished.\n");
    }

    return 0;
}
```

The screenshot shows a Visual Studio Code editor window with the following components:

- EXPLORER:** A file tree on the left showing the project structure under "OPERATING SYSTEM [WSL: UBUNTU-24.04]". The files listed are:
 - Lab03-hometask
 - task01.c
 - task02.c (selected)
 - lab03-task01.c
 - lab03-task02.c
 - lab03-task03.c
 - lab03-task04.c
 - out-task03
 - task01-output
 - task02-output
 - task03-output
 - task04-output
- EDITOR:** The main workspace showing the code for "task02.c". The code is as follows:

```
1 #include <sys/wait.h>
2
3 int main() {
4     pid_t pid = fork(); // create child process
5
6     if (pid < 0) {
7         perror("fork failed");
8         return 1;
9     }
10    else if (pid == 0) {
11        // Child process
12    }
13 }
```
- TERMINAL:** A terminal window at the bottom showing the execution of the program. The output is as follows:

```
afiamaham@DESKTOP-190QPQH:~/Operating System$ gcc Lab03-hometask/task02.c -o out-task03
afiamaham@DESKTOP-190QPQH:~/Operating System$ ./out-task03
Parent (PID: 3635) waiting for child (PID: 3636)...
Child (PID: 3636) running date command...
Tue Oct 7 16:30:59 PKT 2025
Child finished.
afiamaham@DESKTOP-190QPQH:~/Operating System$
```
- STATUS BAR:** At the bottom, it shows "WSL: Ubuntu-24.04" and "Ln 27, Col 1 Spaces: 4 UTF-8 LF {} C".